

ADAM RULE

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EDUCATION

- 9/2013 – 6/2018 University of California, San Diego
PHD, *Cognitive Science*
- 9/2011 – 6/2013 University of Washington, Seattle
MS, *Human Centered Design & Engineering*
- 8/2008 – 5/2011 University of Illinois, Urbana-Champaign
BS, *Industrial Engineering*—HIGHEST HONORS

THESIS

- 6/2018 Design and Use of Computational Notebooks
Advisor: Jim Hollan
Committee: Gloria Mark, Brad Voytek, Bill Griswold, Philip Guo, Robert El-Kareh

ACADEMIC POSITIONS AND WORK EXPERIENCE

- 8/2021 – PRES. Assistant Professor of Information Science
University of Wisconsin–Madison
- 11/2021 – PRES. Honorary Fellow of Family Medicine and Community Health
University of Wisconsin–Madison
- 1/2019 – 6/2021 Postdoctoral Scholar of Medical Informatics and Clinical Epidemiology
Oregon Health & Science University
- 7/2018 – 9/2018 Postdoctoral Scholar of Cognitive Science
UC San Diego
- 9/2013 – 6/2018 Graduate Research Assistant
UC San Diego

AWARDS, HONORS

AWARDS

- 2021 Best Paper, 2nd Place, MedInfo 2021 [C1]
2021 Distinguished Paper Award Nominee, AMIA 2021 – Top 10 papers [C2]
2018 Honorable Mention, CHI 2018 — top 5% of all submissions [C5]
2016 Outstanding Teaching Award, Cognitive Science Department, Winter 2016
2012 Honorable Mention, HRI 2012 — top 5% of all submissions [C8]
2012 Microsoft Design Expo — Best Presentation

FELLOWSHIPS

- 2019 - 2021 National Library of Medicine Postdoctoral Training Grant

2014 - 2017 National Library of Medicine Predoctoral Training Grant — 3 years tuition & stipend

RESEARCH AND PUBLICATIONS

* Peer Reviewed, †Based on research conducted prior to UW-Madison appointment
The leading conferences in the fields of human computer interaction and medical informatics (i.e., AMIA, MedInfo, CHI, CSCW) are peer-reviewed and archival.
Citation rates from Google Scholar on May 26, 2026

Journal Publications

- J1 **Rule, Adam**, Jeffrey J. Baltus, Mark A. Micek, Christine A. Sinsky, and Brian G. Arndt. Variation in Measures of Electronic Health Record Use Outside Scheduled Hours: A Cross-Sectional Study of Academic Primary Care Physicians. *Applied Clinical Informatics*. 2026. 17(2):307-314.* Citations: 0
- J2 **Adam Rule**, Rutvi Shah, Christina Dudley, Mark A Micek, Brign G Arndt. Primary care physicians' experiences with inbox triage. *JAMIA Open*. 2025. Oct;8(5):ooaf105.* Citations: 1
- J3 **Adam Rule**, Phillip Vang, Mark A Micek, Brian G Arndt. Primary care staff members' experiences with managing electronic health record inbox messages. *Journal of the American Medical Informatics Association*. 2025. 32(6):1040-1049.* Citations: 5
- J4 Daniel Tawfik, **Adam Rule**, Aram Alexanian, Dori Cross, A Jay Holmgren, Sunny S Lou, Eugenia McPeck Hinz, Christian Rose, Ratnalekha VN Viswanadham, Rebecca G Mishuris, Jorge M Rodríguez-Fernández, Eric W Ford, Sarah T Florig, Christine A Sinsky, Nate C Apathy. Emerging Domains for Measuring Health Care Delivery With Electronic Health Record Metadata. *Journal of Medical Internet Research*. 2025 Mar 6;27:e64721.* Citations: 6
- J5 **Adam Rule**, Robert El-Kareh. Employing EHR usage data to improve care requires clinician engagement and trust. *Journal Hospital Medicine*. 2025. 20(7):791-792. Citations: 1
- J6 **Adam Rule**, Thomas Kannampallil, Michelle R Hribar, Adam C Dziorny, Robert Thombley, Nate C Apathy, Julia Adler-Milstein. Guidance for Reporting Analysis of Metadata on Electronic Health Record Use. *Journal of the American Medical Informatics Association*. 2024. 31(3):784-789.* Citations: 31
- J7 Brian G Arndt, Mark A Micek, **Adam Rule**, Christina M Shafer, Jeffrey J Baltus, Christine A Sinsky. More Tethered to the EHR: EHR Workload Trends Among Academic Primary Care Physicians, 2019-2023. *Annals of Family Medicine*. 2024. 22(1):12-18.* Citations: 64
- J8 Mark A Micek, **Adam Rule**, Jeffrey J Baltus, Brian G Arndt. Association of Primary Care Physicians' Ambulatory Full-time Equivalencies With Time in the Electronic Health Record. *JAMA Network Open*. 2023. 6(6): e2320032-e2320032.* Citations: 7

- J9 Brian G Arndt, Mark A Micek, **Adam Rule**, Christina M Shafer, Jeffrey J Baltus, Christine A Sinsky. Refining Vendor-Defined Measures to Accurately Quantify EHR Workload Outside Time Scheduled with Patients. *Annals of Family Medicine*. 2023. 21(3): 264-268.* Citations: 19
- J10 **Adam Rule**, Christina M Shafer, Mark A Micek, Jeffrey J Baltus, Christine A Sinsky, Brian G Arndt. Gender Differences in Primary Care Physicians' Electronic Health Record Use over Time: An Observational Study. *Journal of the General Internal Medicine*. 2023.* Citations: 21
- J11 **Adam Rule**, Edward R Melnick, Nate C Apathy. Using event logs to observe interactions with electronic health records: an updated scoping review shows increasing use of vendor-derived measures. *Journal of the American Medical Informatics Association*. 2023. 30(1):144-154* Citations: 58
- J12 **Adam Rule**, Michelle Hribar. Frequent but fragmented: use of note templates to document outpatient visits at an academic health center. *Journal of the American Medical Informatics Association*. 2022. 29(1):137-141** Citations: 29
- J13 Haley Dusek, Isaac Goldstein, **Adam Rule**, Michael Chiang, Michelle Hribar. Clinical Documentation During Scribed and Non-scribed Ophthalmology Office Visits. *Ophthalmology Science*. 2021. 15(1):100088** Citations: 5
- J14 Jimmy Chen, Michelle Hribar, Isaac Goldstein, **Adam Rule**, Wei-Chun Lin, Haley Dusek, Michael Chiang. Electronic health record note review in an outpatient specialty clinic: who is looking? *JAMIA Open*. 2021. 4(3): ooab044** Citations: 4
- J15 **Adam Rule**, Steven Bedrick, Michael Chiang, Michelle Hribar. Length and Redundancy of Outpatient Progress Notes Across a Decade at an Academic Medical Center. *JAMA Network Open*. 2021. 4(7):e2115334-* Citations: 101
- J16 Christine Sinsky, **Adam Rule**, Genna Cohen, Brian Arndt, Tait Shanafelt, Christopher Sharp, Sally Baxter, Ming Tai-Seale, Sherry Yan, You Chen, Julia Adler-Milstein, Michelle Hribar. Metrics for assessing physician activity using electronic health record log data. *Journal of the American Medical Informatics Association*. 2020. 27(4):639-643** Citations: 181
- J17 Brad Henriksen, Isaac Goldstein, **Adam Rule**, Abigail Huang, Haley Dusek, Austin Igelman, Michael Chiang, Michelle Hribar. Electronic Health Records in Ophthalmology: Source and Method of Documentation. *American Journal of Ophthalmology*. 2020. 211:191-9** Citations: 19
- J18 **Adam Rule**, Michelle Hribar, Michael Chiang. Using Electronic Health Record Audit Logs to Study Clinical Activity: A Systematic Review of Aims, Measures, and Methods. *Journal of the American Medical Informatics Association*. 2020. 27(3):480-90** Citations: 154
- J19 Michelle Hribar, **Adam Rule**, Abigail Huang, Haley Dusek, Isaac Goldstein, Bradley Henriksen, Wei-Chun Lin, Austin Igelman, Michael Chiang. Redundancy of Progress Notes for Serial Office Visits. *Ophthalmology*. 2020. 127(1):134** Citations: 9
- J20 **Adam Rule**, Amanda Birmingham, Cristal Zuniga, Ilkay Altintas, Shih-Cheng Huang, Rob Knight, Niema Moshiri, Mai H. Nguyen, Sara Brin Rosenthal, Fernando Pérez, Peter W. Rose. Ten Simple Rules for Writing and Sharing Computational

Analyses in Jupyter Notebooks. *PLoS Computational Biology*. 2019. 15(7): e1007007[†]
Citations: 239

- J21 **Adam Rule**, Ian Drosos, Aurélien Tabard, Jim Hollan. Aiding Collaborative Reuse of Computational Notebooks with Annotated Cell Folding. *Proceedings of the ACM, Human-Computer Interaction*. 2018. 2(CSCW):1-12[†] Citations: 88
- J22 **Adam Rule**, Aurélien Tabard, Jim Hollan. Using Visual Histories to Reconstruct the Mental Context of Suspended Activities. *Human-Computer Interaction*. 2017. 32(5-6):511-58[†] Citations: 27
- J23 Tim F Liao, **Adam Rule**, Ryanne Ardisana, Alexandra Knitcher, Amanda Mayo Corey Sarcu. Social Behavior in Public Spaces in a College Town. *Sociologija i prostor/Sociology & Space*. 2012. 50(1):3-26[†] Citations: 16

Conference Publications

 = Best paper honorable mention

- C1  **Adam Rule**, Michael F Chiang, Michelle R Hribar. Medical Scribes Have a Variable Impact on Documentation Workflows. In *MEDINFO 2021: One World, One Health—Global Partnership for Digital Innovation*, 2021[†] Citations: 5
BEST PAPER, 2ND PLACE
- C2  **Adam Rule**, Sarah Florig , Steven Bedrick , Vishnu Mohan , Jeffrey Gold, Michelle Hribar. Comparing Scribed and Non-scribed Outpatient Progress Notes. In *Proceedings of the American Medical Informatics Association Annual Symposium (AMIA)*, 2021.[†] Citations: 11
DISTINGUISHED PAPER NOMINEE (10 OF 84)
- C3 Michelle Hribar, Haley Dusek, Isaac Goldstein, **Adam Rule**, Michael Chiang. Methods for Large-Scale Quantitative Analysis of Scribe Impacts on Clinical Documentation. In *Proceedings of the American Medical Informatics Association Annual Symposium (AMIA)*, 2020[†] Citations: 7
- C4 **Adam Rule**, Isaac Goldstein, Michelle Hribar, Michael Chiang. Clinical Documentation as End-User Programming. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*, 2020[†] Citations: 19
- C5  **Adam Rule**, Aurélien Tabard, Jim Hollan. Exploration and Explanation in Computational Notebooks. In *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI)*, 2018[†] Citations: 502
HONORABLE MENTION, TOP 5% OF SUBMISSIONS
- C6 **Adam Rule**, Steven Rick, Michael Chiu, Phillip Rios, Shazia Ashfaq, Alan Calvitti, Wesley Chan, Nadir Weibel, Zia Agha. Validating free-text order entry for a note-centric EHR. In *Proceedings of the American Medical Informatics Association Annual Meeting (AMIA)*, 2015[†] Citations: 13
- C7 **Adam Rule**, Aurélien Tabard, Karen Boyd, Jim Hollan. Restoring the Context of Interrupted Work with Desktop Thumbnails. In *Proceedings of the Cognitive Science Society Annual Meeting (CogSci)*, 2015[†] Citations: 15

- C8 **Adam Rule**, Jodi Forlizzi. Designing interfaces for multi-user, multi-robot systems. In *Proceedings of the ACM Conference on Human Robot Interaction (HRI)*, 2012** Citations: 33
-  **HONORABLE MENTION, TOP 5% OF SUBMISSIONS**

Doctoral Consortium, Workshops, Posters & Abstracts

- A1 Alan Labrum, Kathryn Marxen-Simonson, Tosha Zaback, **Adam Rule**, John Boyer, Mitch Brinks. Six Feet of Separation: Physical Distancing Impacts on Blind and Low Vision During COVID-19 Pandemic. *Investigative Ophthalmology & Visual Science (ARVO)*, 2021** Citations: 4
- A2 Michelle Hribar, Haley Dusek, Isaac Goldstein, **Adam Rule**, Michael Chiang. Documentation Composition and Efficiency During Scribed and Non-Scribed Ophthalmology Visits. *Investigative Ophthalmology & Visual Science (ARVO)*, 2020** Citations: 1
- A3 Sarah Glass, **Adam Rule**, Michelle Hribar, Thomas Hwang, Michael Chiang. Electronic Health Records in Ophthalmology: Quantitative evaluation of progress note utility for clinical decision-making. *Investigative Ophthalmology & Visual Science (ARVO)*, 2020** Citations: 0
- W1 **Adam Rule**, Aurélien Tabard, Jim Hollan. Traces: A Flexible, Open Source Activity Tracker for Workplace Studies. *Workshop on the Quantified Workplace at the ACM Conference on Computer Supported Cooperative Work (CSCW)*, 2017** Citations: 0
- P1 **Adam Rule**, Erin Gillespie, Nadir Weibel, Todd Pawlicki. Design Thinking in Radiation Oncology. *Human Factors and Ergonomics Society Healthcare Symposium*, 2016** Citations: 0
- P2 **Adam Rule**, Aurélien Tabard, Jim Hollan. Thinking in 4D: Preserving and Sharing Mental Context Across Time. In *Proceedings of the ACM Conference on Computer Supported Cooperative Work and Social Computing Companion (CSCW)*, 2016** Citations: 5
- P3 Karen Boyd, **Adam Rule**, Aurélien Tabard, Jim Hollan. Human Values in Tracking and Sharing Computer-Mediated Activity at Work. In *Proceedings of the ACM Conference on Computer Supported Cooperative Work and Social Computing Companion (CSCW)*, 2016** Citations: 10
- D1 **Adam Rule**. Visualizing Computer Activity to Support the Resumption of Long-term Creative Work. In *Proceedings of the SIGCHI Conference on Creativity & Cognition*, 2015** Citations: 5

RESEARCH AND PUBLICATIONS IN PROGRESS

- IP1 **Adam Rule**, Yongseong Kim, Ajil Joseph. Jeff Baltus, Mark Micek, Chris Sinsky, Brian Arndt. Generative Artificial Intelligence in Primary Care: A Review of Benefits, Risks, and Interaction Paradigms as Described by Clinicians, Staff, Administrators, and Technologists in News Media. In *Proceedings of the American Medical Informatics Association Annual Symposium (AMIA)* 2026. (Accepted)

RESEARCH SUPPORT

EXTRAMURAL

- 6/2023 - 5/2024 Characterizing Team Support for Medication Orders in Primary Care. American Medical Association.
Role: Co-Investigator. PI: Brian Arndt
(\$59,987)
- 5/2022 - 4/2023 Off the Clock: Validating Measures of After-hours EHR Work in Primary Care at UW Health. American Medical Association.
Role: Co-Investigator. PI: Brian Arndt
(\$49,964)

INTRAMURAL

- 7/2024 - 6/2026 Characterizing Team Support for Medication Orders in Primary Care. Department of Family Medicine and Community Health Small Research Grant.
Role: PI.
(\$24,107)

PRESENTATIONS

* Peer Reviewed

Conference Presentations

- P1 Adam Rule, Phillip Vang, Mark A Micek, Brian G Arndt. Primary care staff members' experiences with managing electronic health record inbox messages. *Symposium of the American Medical Informatics Association (AMIA)*. 2025, Nov 15-19. Atlanta, GA.*
- P2 Adam Rule, Rutvi Shah, Christina Dudley, Mark A Micek, Brian G Arndt. Triaging Inbox Work: An Interview Study with Primary Care Physicians. *Symposium of the American Medical Informatics Association (AMIA)*. 2024, Nov 9-13. San Francisco, CA.*
- P3 Adam Rule, Jeffrey J Baltus, Christine A Sinsky, Mark A Micek, Brian G Arndt. Variation in Measures of EHR Use Outside Scheduled Clinic Hours. *Symposium of the American Medical Informatics Association (AMIA)*. 2023, Nov 11-15. New Orleans, LA.*

Invited Presentations

Cognitive Science and Human Computer Interaction
UW Health, Clinical Informatics Fellowship Seminar, Oct 2025
UW Health, Clinical Informatics Fellowship Seminar, Sept 2024
UW Health, Clinical Informatics Fellowship Seminar, Sept 2023

Primary Care Staff Members Experiences with Managing Electronic Health Record Inbox Messages
JAMIA Journal Club, Aug 2025

Experiences with Inbox Work Among Primary Care Physicians and Staff
General Internal Medicine Conference, UW-Madison, Apr 2025

Communicating with Your Doctor in the Age of AI

UW-Madison, LSC 617: Health Communication in the Information Age, Oct 2024

A Brief Introduction to (and History of) EHR Use Measures

Clinical Informatics Program Directors Didactic Series, Sept 2024

Guidance for Reporting Analyses of Metadata on Electronic Health Record Use

National Research Network for EHR Audit Log Data, Jan 2024

Composition of Clinical Notes

UW-Madison, LSC 617: Health Communication in the Information Age, Nov 2023

UW-Madison, COM ARTS 617: Health Communication in the Information Age, Nov 2022

University of Wisconsin-Madison, LIS 201, The Information Society, Oct 2022

UW Health, Medical Informatics Group, Feb 2022

UC San Diego, Ophthalmic Informatics Group, Jan 2022

University of Wisconsin-Madison, CIBM Seminar, Oct 2021

Oregon Health & Science University, Casey Eye Institute, Jun 2019

Electronic Health Record Use in Primary Care

WisconsInformatics, Jan 2024

University of Wisconsin-Madison, Population Health Seminar, Dec 2022

UW Health, Clinical Informatics Community of Practice, Oct 2022

Electronic Health Records

University of Wisconsin-Madison, CS 202, Intro to Computer Science, Jul 2022

Using Software Logs to Study Electronic Health Record Use in Primary Care

University of Wisconsin-Madison, LIS 613: User Experience Design III, Jun 2022

University of Wisconsin-Madison, Biostatistics and Medical Informatics Seminar, Mar 2022

The Design and Use of Computational Notebooks

University of Washington, Interactive Data Lab, Dec 2018

Oregon Health & Science University, Medical Informatics, Nov 2018

Field Methods in Human Computer Interaction

UC San Diego, COGS 13: Field Methods, May 2018

Exploration and Explanation in Computational Notebooks

UC Berkeley, Data Science Institute, Dec 2017

Cognitive Science & Design

UC San Diego, COGS 1: Intro to Cognitive Science, Nov 2016

UC San Diego, COGS 1: Intro to Cognitive Science, Oct 2015

TEACHING EXPERIENCE

UW-Madison, Instructor

Spring 2025 LIS 612 — User Experience Design II

Spring 2025 LIS 612 — User Experience Design II (Online)

Fall 2024 LIS 613 — User Experience Design III

Fall 2024 LIS 615 — Systems Analysis and Project Management for Information Professionals

Spring 2024 LIS 612 — User Experience Design II

Fall 2023 LIS 613 — User Experience Design III

Fall 2023	LIS 615 — Systems Analysis and Project Management for Information Professionals
Spring 2023	LIS 612 — User Experience Design II
Spring 2023	LIS 470 — Interaction Design Studio
Fall 2022	LIS 613 — User Experience Design III
Fall 2022	LIS 615 — Systems Analysis and Project Management for Information Professionals
Spring 2022	LIS 612 — User Experience Design II
Fall 2021	LIS 341 — Intro to Interaction Design
Fall 2021	LIS 615 — Systems Analysis and Project Management for Information Professionals
	<i>Oregon Health & Science University, Teaching Assistant</i>
Summer 2019	BMI 510 — Human Computer Interaction in Biomedicine
	<i>UC San Diego, Teaching Assistant</i>
Spring 2018	COGS 10 — Cognitive Consequences of Technology
Winter 2017	DSGN 1 — Design of Everyday Things
Winter 2016	CSE 170/COGS 120 — HCI Design - Outstanding Teaching Award
Winter 2015	COGS 102b — Cognitive Ethnography
Spring 2014	COGS 102c — Cognitive Design Studio

SERVICE

DEPARTMENTAL

2025-2026	PhD Committee
2024-2025	Faculty Hiring Committee
2024-2025	Curriculum Committee, MS Info
2021-2024	PhD Committee

REVIEWING

CHI (2016, 19-24), CSCW (2019, 21, 22), IUI (2019), UIST (2020) AMIA (2015-16, 19-23), JAMIA (2020-24), JAMIA Open (2020), JBI (2020), Ophthalmology (2019), Applied Clinical Informatics (2019, 21-24), JAMA Network Open (2021, 2024), International Journal of Medical Informatics (2022), Family Medicine (2023, 2024), Journal of Hospital Medicine (2024, 2026), Ophthalmology Science (2024), Journal of Medical Internet Research (2024)

SPECIAL RECOGNITION FOR OUTSTANDING REVIEWS

CHI (2020, 2021, 2025, 2026), CSCW (2020, 2022), Family Medicine (Top Reviewer, 2024)

OTHER SERVICE

2026	Review Panelist, National Science Foundation
2019	AMIA Student Paper Competition Committee
2019	EXITO Career Panel — Research Diversity program at Portland State University